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Intelligent Horizontal Well Placement Optimization: Modeling and Application – A Case Study of Low Permeability Carbonate Reservoir in the Middle East

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Abstract

Horizontal wells are widely used in the development of carbonate reservoirs in the Middle East, and optimizing their placement is crucial for efficient and cost-effective production. This study focuses on creating an intelligent optimization model for horizontal well placement in a carbonate reservoir, aiming to enhance oil recovery and maximize economic benefits. By integrating intelligent optimization algorithms with advanced reservoir simulations, the study addresses the challenges posed by the heterogeneity and complex geological characteristics of carbonate formations. The proposed workflow combines the Weighted Mapping of Productivity Potential (WMPP) method with the grey wolf optimizer (GWO) to optimize well trajectories. WMPP incorporates key reservoir parameters like porosity, permeability, and fluid saturation to generate productivity maps, while GWO is used to determine the optimal weighting coefficients for WMPP, ensuring precise identification of high-productivity zones and avoiding areas of high water saturation or geological uncertainty. A representative carbonate reservoir model was developed using geological, petrophysical, and dynamic data to validate the methodology. The WMPP-guided placement was compared to conventional oil initially in place (OOIP)-based methods, with a focus on production performance, efficiency, and economic outcomes. The study, applied to a low-permeability carbonate reservoir in the Middle East, demonstrated the effectiveness of WMPP-guided horizontal well placement in improving reservoir performance. Compared to the OOIP approach, the WMPP method required fewer wells while increasing cumulative oil production over a 20-year period. The optimization results highlighted the WMPP technique's ability to navigate the complex, multimodal objective function landscape of carbonate reservoirs, ensuring maximum reservoir contact with high-permeability zones, minimizing water breakthrough, and reducing drilling costs. This case study illustrates the importance of integrating

advanced optimization algorithms with reservoir simulations to enhance production efficiency and resource utilization. Ultimately, this study presents a novel application of WMPP and grey wolf optimizer for optimizing horizontal well placement in carbonate reservoirs, providing a scalable framework applicable to similar reservoirs worldwide and contributing to the advancement of petroleum engineering knowledge.

Key words: Intelligent Horizontal Well Placement Optimization, Modeling and Application, Case Study, Middle East

Introduction

The optimization of well placement in carbonate reservoirs is a critical aspect of enhancing oil recovery and maximizing economic returns (Mogensen and Masalmeh, 2020; Davidescu et al., 2023). In regions like the Middle East, where carbonate reservoirs are abundant, the challenge is particularly pronounced due to the complex geological structures and significant heterogeneity of these reservoirs (Buza, 2008). The variation in porosity, permeability, fluid saturation, and other petrophysical properties leads to unpredictable flow behaviors (Feng et al., 2018 and 2021; Zhang et al., 2023; Zhang et al., 2024), making it difficult to optimize well placement effectively. Horizontal drilling, which can improve reservoir contact, enhance production rates, and minimize water breakthrough, presents a promising solution (Bennion et al., 1996). However, the strategic placement of horizontal wells in such heterogeneous formations requires sophisticated optimization methods.

Traditionally, well placement has relied on analytical methods, integrating reservoir physics and geo-mechanical theory with simulation-calibrated empirical frameworks. (Arinkoola et al., 2016). Analytical approaches, employing permeability resistance equivalence and directional well arrays, these approaches refine well placement by enhancing breakthrough intervals and volumetric displacement efficiency. (Manrique et al., 2012; Pirayesh et al., 2014). Semi-empirical methods, on the other hand, related to formulating well placement schemes according to knowledge and comparing them through reservoir simulations (Waheed et al., 2001). However, these approaches often face limitations, including subjective influence and difficulties in achieving the truly optimal solution.

With advancements in computing and intelligent oilfield technologies (Zhang et al., 2023; Wang et al., 2022 and 2025), the integration of optimization theory with reservoir simulation has transformed well placement from a complex engineering challenge into a feasible optimization problem (Zhao et al., 2020). Techniques like genetic algorithms and surrogate models have been applied to improve placement (Wang et al., 2022). In particular, the grey wolf optimizer (GWO) algorithm (Mirjalili et al., 2014), which mimics the hunting behavior of wolves, offers a powerful method for solving non-linear, complex optimization problems such as well placement. GWO's ability to search large solution spaces and its efficiency in handling multimodal problems make it particularly well-suited for optimizing horizontal well placement in carbonate reservoirs.

The WMPP is a critical technique in optimizing well placement in heterogeneous reservoirs (Guerra and Narayanasamy, 2006). WMPP maps regions of varying productivity potential within the reservoir by incorporating key indexes such as saturation, pressure, permeability, and distance to boundaries. However, the traditional direct mapping of productivity potential (DMPP) often fails to account for reservoir heterogeneity and dynamic fluid behavior (Ding et al., 2014). To address this, the WMPP method introduces trainable weights that allow the model to adapt to the specific geological and physical characteristics of the reservoir, improving the accuracy and effectiveness of the well placement strategy.

This study focuses on developing an intelligent optimization model for horizontal well placement in carbonate reservoirs, utilizing the Grey Wolf Optimizer (GWO) to enhance well placement and maximize oil recovery and economic benefits. The model combines the WMPP with advanced reservoir simulation techniques to navigate the complexities of carbonate formations. By harnessing the optimization power of GWO, the study aims to improve production efficiency, reduce the number of wells, and lower drilling

costs. Traditional methods often struggle with the challenges posed by reservoir heterogeneity, geological uncertainty, and fluid distribution. In contrast, the integration of GWO with WMPP provides a more data-driven and robust approach for optimizing well placement in such complex environments. The findings are expected to offer valuable insights for reservoir management, creating a scalable framework for optimizing well placement in carbonate reservoirs, particularly in regions like the Middle East.

Related work

This section will provide an in-depth discussion of the algorithms used in this research and the workflow of the well placement optimization method. First, the WMPP method, which is derived from the DMPP method, will be introduced. Next, the objective function for the optimization of the well placement will be introduced. Lastly, the workflow of the grey wolf optimizer (GWO), employed to solve the WMPP indices, will be presented.

Weighted Mapping of Productivity Potential

The DMPP is a method used to represent regions of varying production potential within a reservoir. It is mathematically expressed as:

$$J_{(i,j,k)}(t) = [S_{o,i,j,k}(t) - Sor] \cdot [p_{o,i,j,k}(t) - p_{min}] \cdot \ln(K_{i,j,k}) \cdot \ln(r_{i,j,k}) \cdot d_{woc,i,j,k} \cdot \phi_{i,j,k} \cdot d_{goc,i,j,k} \quad (1)$$

Where: i, j, k represent the x, y, z directions; $J_{(i,j,k)}(t)$ represents the DMPP at grid block (i, j, k) at time t ; $S_{o,i,j,k}(t)$ represents the oil saturation; Sor is the residual oil saturation at the grid block; $p_{o,i,j,k}(t)$ represents the oil pressure; p_{min} is the minimum bottomhole pressure; $K_{i,j,k}$ denotes permeability at the grid block; $r_{i,j,k}$ is the distance to the closest boundary; $d_{woc,i,j,k}$ represents the distance to the nearest oil-water contact; $\phi_{i,j,k}$ represents the porosity; $d_{goc,i,j,k}$ represents the distance.

The DMPP is a basic and direct approach that simplifies the reservoir's complexities but does not account for varying geological and physical characteristics, nor does it incorporate production schedules. While it offers a general framework, its lack of adaptability to diverse reservoirs presents a limitation.

To overcome these shortcomings, we utilize the WMPP in this study. The WMPP improves upon the DMPP by incorporating adjustable weights, facilitating process refinement aligned with target reservoir dynamics. This enhancement makes the WMPP more adaptable and valuable for reservoirs with varying characteristics. Mathematically, the WMPP is given by:

$$I_{(i,j,k)}(t) = \omega_1 [S_{o,i,j,k}(t) - Sor] + \omega_2 [p_{o,i,j,k}(t) - p_{min}] + \omega_3 \ln(K_{i,j,k}) + \omega_4 \ln(r_{i,j,k}) + \omega_5 d_{woc,i,j,k} + \omega_6 \phi_{i,j,k} + \omega_7 d_{goc,i,j,k} \quad (2)$$

Where: $I_{(i,j,k)}(t)$ represents the WMPP at the grid block (i, j, k) at time t , and ω is a vector of weights:

$$\omega = [\omega_1, \omega_2, \dots, \omega_7] \quad (3)$$

These trainable weights allow the WMPP to be customized to the unique characteristics of a reservoir, enabling the model to adapt to specific geological, physical, and fluid properties. This flexibility ensures optimal well placement by adjusting to evolving reservoir conditions, making it a more accurate and effective tool for maximizing production and improving economic outcomes.

Objective function

Grid blocks traversed by well trajectories form the basis for WMPP normalization. Each well's WMPP is computed through spatial averaging of grid values contained in its well cut region, formulated as:

$$\text{WMPP}_w(\omega) = \omega \left(\frac{\sum_{g_{w,n}=1}^{g_{w,N}} \mathbf{x}_{g_{w,n}}}{g_{w,N}} \right) \quad (4)$$

Where: $g_{w,N}$ is the total number of grids in the well cut, and $\mathbf{x}_{g_{w,n}}$ is the WMPP for grid $g_{w,n}$.

The vector $(\text{WMPP}_w, \text{WOPT}_w)$, which are used to optimize the weight vector ω , ensuring that wells with higher WMPP_w generally show higher WOPT_w . Since WMPP_w is linear and WOPT_w involves complex, nonlinear functions that cannot directly predict WOPT_w from WMPP_w .

Instead, the Spearman rank correlation coefficient is utilized to model the monotonic relationship between WMPP_w and WOPT_w , formulated as:

$$\rho(\text{WMPP}_w, \text{WOPT}_w) = \frac{\text{cov}(R(\text{WMPP}_w), R(\text{WOPT}_w))}{\sigma(\text{WMPP}_w)\sigma(\text{WOPT}_w)} \quad (5)$$

The optimization problem is then:

$$\omega^* = \text{argmax} \left(\frac{\text{cov}(R(\text{WMPP}_w), R(\text{WOPT}_w))}{\sigma(\text{WMPP}_w)\sigma(\text{WOPT}_w)} \right) \quad (6)$$

The goal is to maximize the Spearman correlation, ensuring a reliable reference for well placement decisions. The following sections detail the method for obtaining the optimized ω^* using gray wolf optimizer.

Grew Wolf Algorithm

The Gray Wolf Optimizer (GWO) is a nature-inspired optimization algorithm based on the hunting behavior of gray wolves as shown in Fig. 1. It mimics the social hierarchy and hunting strategies of a wolf pack to find optimal solutions to complex optimization problems. The algorithm was introduced by S. Mirjalili et al. in 2014 and has gained popularity due to its simplicity and efficiency in solving continuous and discrete optimization problems. The following equations describe the specific theories of how wolves encircle, hunt, and attack prey. Let α , β , and γ represent top wolves which guide the others during the optimization process. The encircling prey by wolves is illustrated in equations 7 and 8 as follows:

$$\vec{D} = |\vec{C} \cdot \vec{P}_p(t) - \vec{P}(t)| \quad (7)$$

$$\vec{P}(t+1) = \vec{P}_p(t+1) - \vec{A} \cdot \vec{D} \quad (8)$$

where t represents the index of iteration; D represents the intermediate vector; $\vec{P}_p(t)$ and $\vec{P}$ represent the position vector of the prey and wolf, respectively; $\vec{A}$ and $\vec{C}$ represent the coefficient vectors and are calculated as follows:

$$\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a} \quad (9)$$

$$\vec{C} = 2 \cdot \vec{r}_2 \quad (10)$$

where: the elements in vector $\vec{a}$ linearly decrease from two to zero; the elements in vectors $\vec{r}_1$ and $\vec{r}_2$ are randomly generated in the interval from zero to one. The rules of hunting and attacking a prey (preserving the best wolves and renewing the other solutions) are expressed in equations 11, 12, and 13.

$$\vec{D}_\alpha = |\vec{C}_1 \cdot \vec{P}_\alpha - \vec{P}| \quad \vec{D}_\beta = |\vec{C}_2 \cdot \vec{P}_\beta - \vec{P}| \quad \vec{D}_\gamma = |\vec{C}_3 \cdot \vec{P}_\gamma - \vec{P}| \quad (11)$$

$$\vec{P}_1 = \vec{P}_\alpha(t) - \vec{A}_1 \cdot \vec{D}_\alpha \quad \vec{P}_2 = \vec{P}_\beta(t) - \vec{A}_2 \cdot \vec{D}_\beta \quad \vec{P}_3 = \vec{P}_\gamma(t) - \vec{A}_3 \cdot \vec{D}_\gamma \quad (12)$$

$$\vec{P}(t+1) = \frac{\vec{P}_1 + \vec{P}_2 + \vec{P}_3}{3} \quad (13)$$

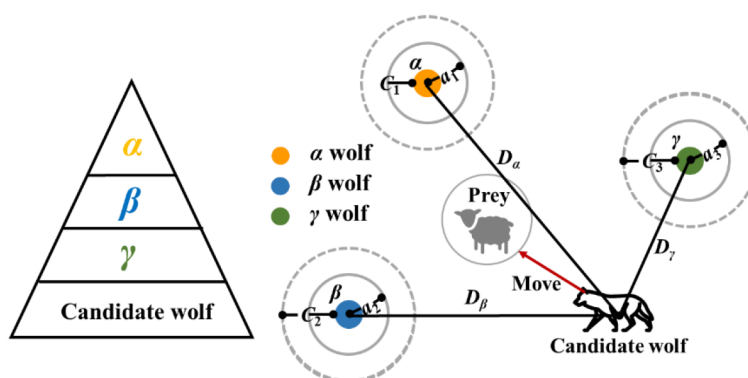


Figure 1—The illustration of GWA (Wang et al., 2023)

In each iteration, the top three solutions are assigned as the α , β , and γ wolves, leading the optimization process. The other δ wolves are updated in the next iteration, allowing for continuous refinement and improvement toward the optimal solution.

In the GWA workflow for reservoir production optimization, shown in Fig. 2, the α , β , and γ wolves represent the top three solutions. The α wolf is the best solution, the β wolf is the second-best, and the γ wolf is the third best. These wolves lead the optimization process, guiding the search for better solutions. The α wolf takes the lead by setting the direction, while the β and γ wolves refine the search, helping avoid local optima. Together, they explore the solution space and continuously adjust their positions based on their fitness values. As the process continues, they drive the system toward increasingly better production schemes. Once the optimization converges—indicating that the solutions have stabilized—the production scheme associated with the α wolf is chosen as the final optimal solution. This solution provides the best configuration for reservoir production, ensuring efficiency and sustainability.

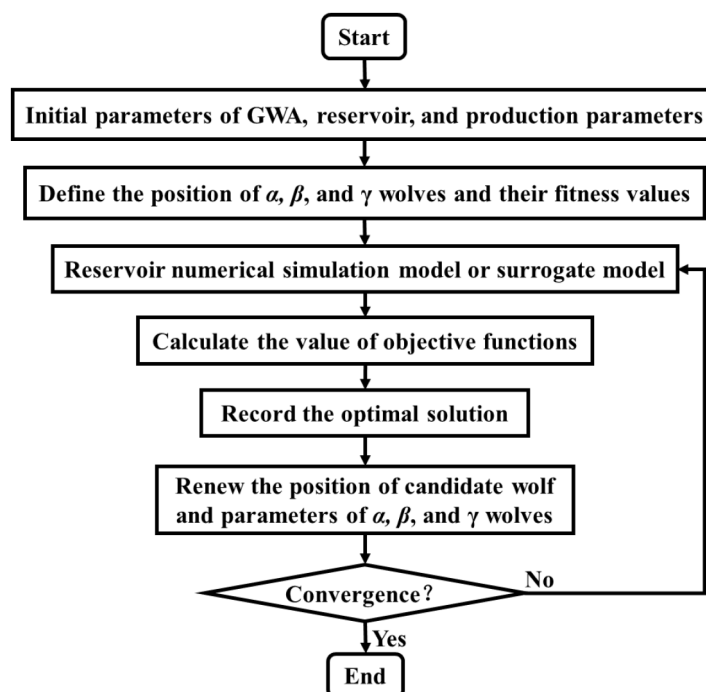


Figure 2—Workflow of GWA algorithm for reservoir production optimization

Workflow

The Weighted Mapping of Productivity Potential (WMPP) method has been developed specifically for optimizing well placement in an oilfield, integrating the results of reservoir simulation model with intelligent algorithms. This method incorporates key spatial properties, such as permeability, porosity, pressure and oil saturation, which are critical factors influencing well performance. WMPP offers a practical, data-driven approach to guide well placement decisions effectively, ensuring that they align with areas that have the highest production potential. The process involves calculating the weighted average of spatial properties around potential well locations, thereby characterizing the average spatial attributes of these sites. This approach accounts for the complex, non-linear relationships that exist between the average spatial attributes of well locations and cumulative production over time, enabling a more accurate prediction of well productivity.

Fig. 3 encapsulates the WMPP computation workflow, involves several key steps. First, numerical simulations are conducted for a range of random well locations, simulating different production scenarios. Next, the average spatial attributes for each well location are computed, considering factors like the surrounding reservoir characteristics. The final step involves the use of the GWA to optimize the weightings assigned to each spatial attribute, fine-tuning the model to achieve the best possible correlation with actual well productivity. Through utilizing the Grey Wolf Algorithm, WMPP optimizes the selection of spatial attributes that most influence production, thereby improving well placement strategies. This results in better decision-making, ensuring that wells are drilled in areas that maximize oil recovery while minimizing operational costs. In addition, the approach enhances overall production efficiency by minimizing well count while enhancing per-well ultimate recovery.

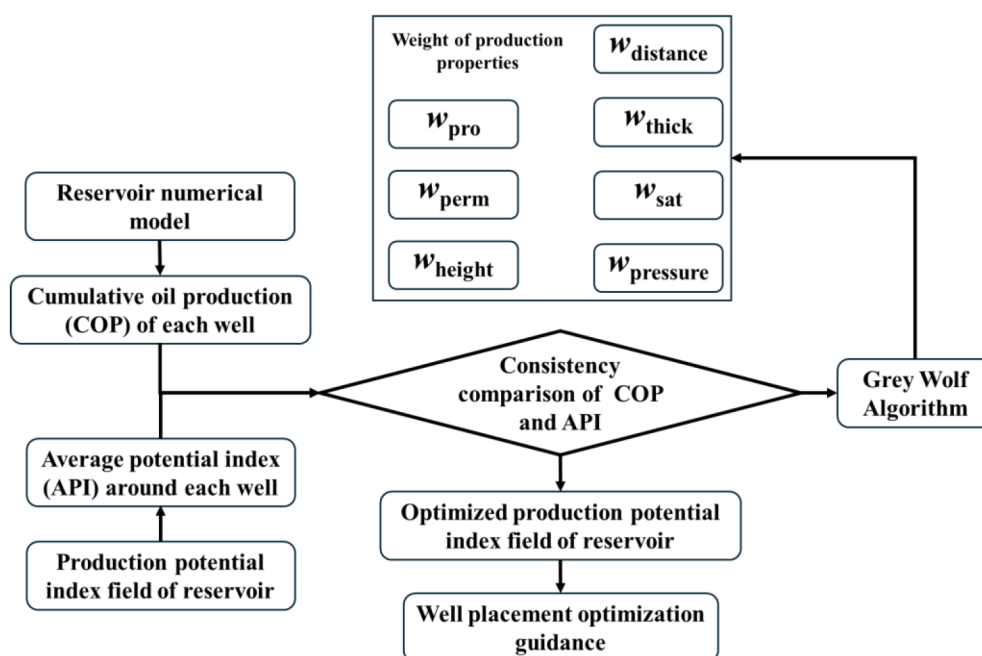


Figure 3—Workflow of WMPP method

Case study

To applicate and validate the reliability of the established well placement optimization method, the Th reservoir unit which represents a practical carbonate reservoir located in the Middle East, offering a detailed simulation of its behavior over time. It is a comprehensive reservoir numerical simulation model that consists of 80 distinct sub-layers, each reflecting a specific geological formation. The reservoir is geologically constrained by faults on the eastern and southern boundaries, influencing its flow and

production characteristics. The model covers $850 \times 320 \times 80$ grid blocks, where each block is 30 meters wide and 30 meters long in the horizontal directions, and 1 meter in the vertical direction.

The permeability and porosity distribution within the reservoir is key to understanding its fluid flow characteristics, as shown in Fig. 4. On average, the porosity of the reservoir is 0.18, while the permeability is 21 millidarcies (mD), providing a general idea of the reservoir's ability to transmit fluids. Importantly, the model does not incorporate an aquifer, which could otherwise affect waterflooding and production behavior during recovery processes.

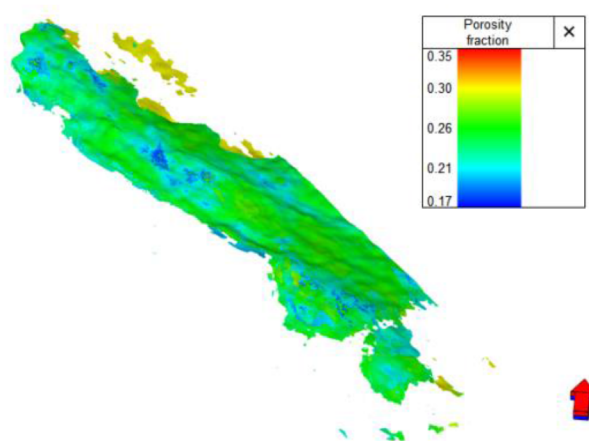


Figure 4—Permeability distribution of Th reservoir

The production scheme is twenty years, during which the reservoir will undergo production at a constant 135 bar bottom hole pressure of producers. This pressure is fixed to maintain well stability and optimize production. In the forthcoming study, which focuses on optimizing well placement, this Th reservoir uses as the base for all simulations and experiments, providing the necessary data to evaluate the most efficient well placement strategies for maximizing recovery from the reservoir. The outcomes of this study will be instrumental in enhancing reservoir management practices for similar geological formations.

A thorough investigation of WOPT-WMPP interactions within the Th model was conducted across 19 optimization cases to establish optimal weighting factors. Subsequent validation via Case-20 examined the coherence between WOPT and three key metrics: WMPP, DMPP, and OOIP (Oil Initially in Place). Spearman correlation analysis encompassing 20 cases quantified interdependencies among WOPT, wellbore parameters, WMPP, and DMPP, confirming the methodology's robustness and practical utility.

Inter-well correlations between WOPT and WMPP production metrics, measured via Spearman's method across 19 cases, are depicted in Fig. 5. The coefficient for CASE12 is the smallest at 0.78, while the average coefficient across all cases is 0.85. This indicates a generally strong relationship between WOPT and WMPP, reflecting the effectiveness of the optimized WMPP field in guiding well placement decisions. A high correlation suggests that the weighting factors applied to WMPP significantly enhance its ability to predict well performance, especially in maximizing cumulative oil production. The close alignment between WOPT and WMPP implies that the optimization approach, based on the weighted mapping of productivity potential, is a reliable tool for guiding well placement strategies. This analysis reinforces the notion that wellbore placement, when informed by the optimized WMPP, can lead to more efficient reservoir exploitation and improved oil recovery outcomes.

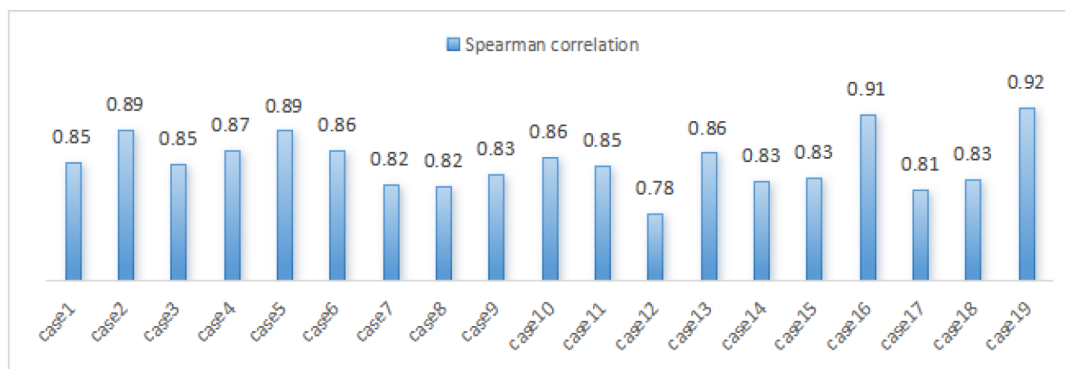


Figure 5—Spearman Correlation between WMPP and WOPT of wells of cases

In addition, the case-20 was developed, featuring 18 producers. The well placement distribution, along with the optimized WMPP, is illustrated in Fig. 6. These cases follow similar operational regime, ensuring consistency in the evaluation process. The cumulative oil production curves for the 18 production wells are shown in Fig. 7, highlighting the production performance over time. The cumulative production values for the wells range from 0.04 to 2.1 million stock tank barrels (MMSTB). This uniformity suggests that the well placements, guided by the optimized WMPP, result in balanced and effective exploitation of the reservoir, maximizing oil recovery while minimizing the risk of localized inefficiencies. The results provide a valuable insight into the effectiveness of the optimized well placement strategy and serve as a demonstration of the potential benefits of integrating advanced optimization techniques in reservoir management. The variation in cumulative production across the wells also reflects the inherent complexity of the reservoir and highlights the importance of intelligent well placement in achieving optimal production outcomes.

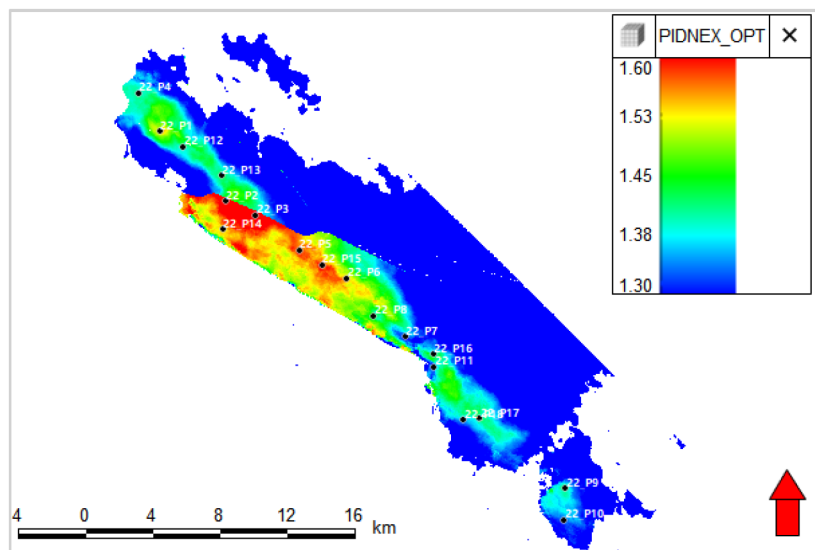


Figure 6—Well placement of CASE-20 and distribution of WMPP

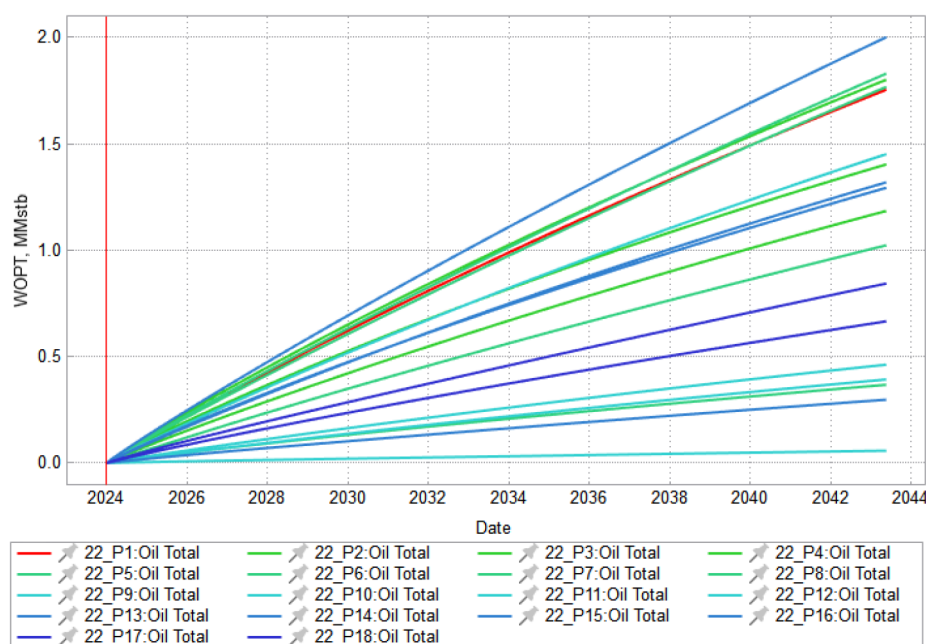


Figure 7—WOPT of wells of CASE-20 vs Date

Field development planning frequently relies on OOIP as a basis for well placement. Fig. 8 illustrates the comparative analysis across all 20 cases. It is apparent that the WMPP yields superior results to OOIP-guided well placement, showing a clear competitive edge. When compared with the documented DMPP technique, WMPP application exhibits consistent and significant improvements overall, with particularly pronounced benefits observed in CASE12 and CASE20. Correlation performance analysis reveals WMPP enhances correlation by 4.9% versus DMPP and achieves a substantial 21.4% improvement over the OOIP method.

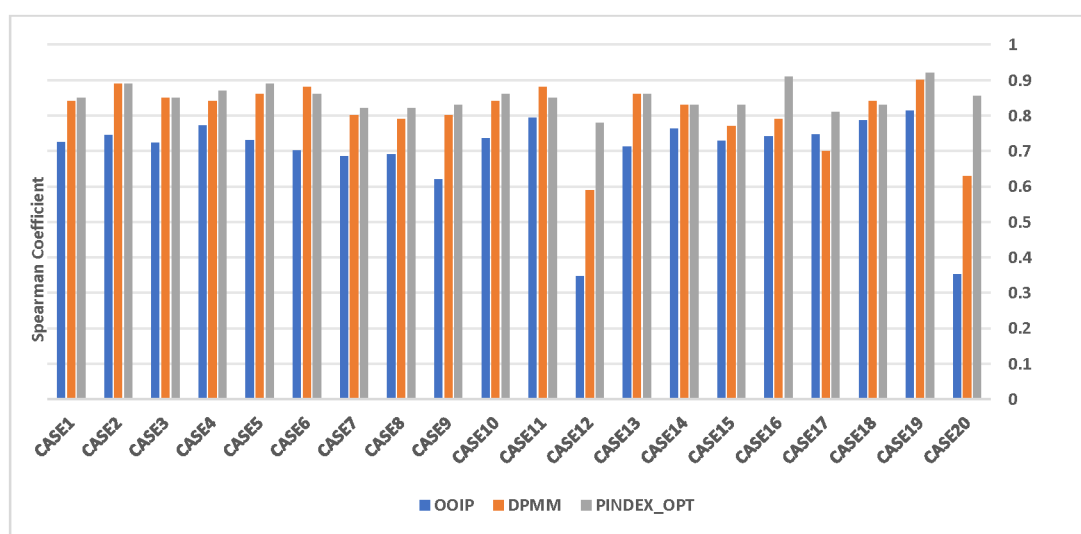


Figure 8—Comparison of OOIP, DPMM and WMPP among cases

Currently, OOIP-based well location optimization constitutes a foundational approach. Fig. 8 offers a comparative visual analysis of all twenty cases. The WMPP method developed surpasses OOIP-based well placement, showcasing the notable improvement. Additionally, when compared to the DMPP approach

discussed before, WMPP outperforms baselines in all scenarios, demonstrating exceptional efficacy in Case 12 and Case-20.

For horizontal well placement optimization implementing the WMPP methodology, scenario A comprised 94 wells positioned using WMPP guidance, whereas scenario B incorporated 102 wells deployed through OOIP-based targeting. Fig. 9 shows the reservoir's WMPP-driven horizontal well configuration. Both development scenarios maintained equivalent injection/production parameters, with producers sustaining 105 bar bottom-hole pressure and field-wide voidage replacement maintained at unity. Simulations spanned the twenty years extraction timeline.

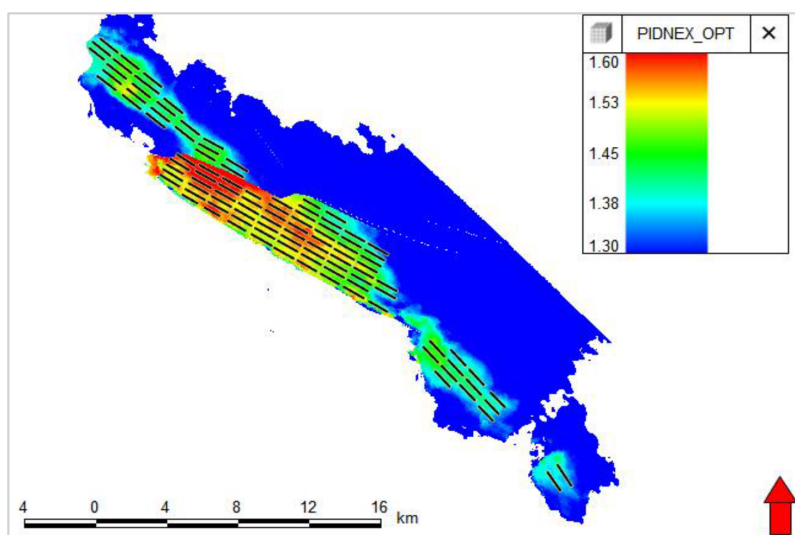


Figure 9—Placement of horizontal wells based on WMPP

Comparative outcomes of both well deployment strategies appear in Fig. 10. The WMPP-driven configuration required 8 fewer wells than the OOIP-based approach yet delivered substantially greater recovery over twenty years production. The oil production volumes increased by 22.5 million stock tank barrels (MMSTB), with average per-well recovery rising from 1.24 to 1.58 MMSTB – a 27.19% enhancement. These metrics confirm the material advantage of WMPP-based development planning.

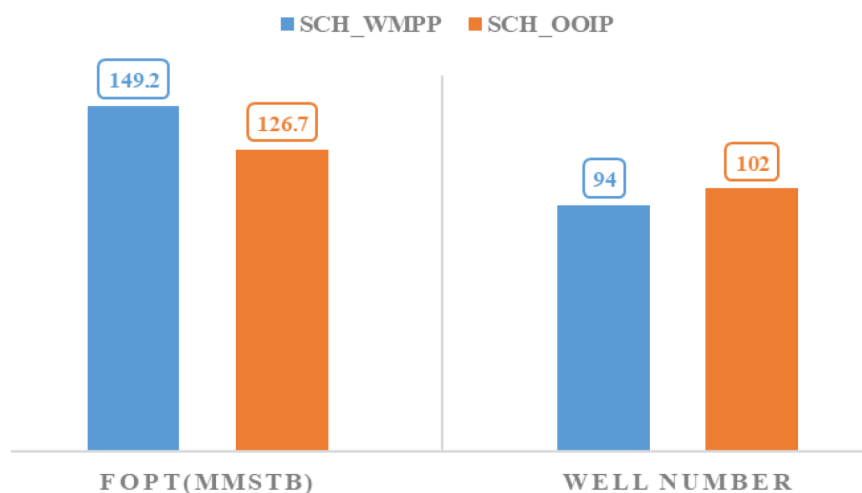


Figure 10—Comparison of FOPT and well numbers of plans guided by WMPP and OOIP

Conclusion

In this study, we introduced and demonstrated the application of the WMPP method to optimize well placement in complex reservoir environments. Through the combination of numerical simulation, advanced algorithms, and machine learning principles, WMPP offers a tailored and efficient solution for guiding well placement decisions. This methodology effectively addresses the challenges posed by complex geological structures while also mitigating the computational complexities typically associated with reservoir well placement optimization.

The grey wolf optimization algorithm enabled identification of ideal weighting coefficients for the WMPP metric, substantially enhancing its Spearman rank correlation with well output (WOPT). Analyses confirmed WMPP's consistent robustness and adaptability across diverse reservoir conditions, exhibiting significant positive correlations with production rates. Comparative evaluations revealed WMPP's marked superiority over conventional approaches—Direct productivity Mapping (DMP) and Original Oil-In-Place (OOIP)—in optimizing well placement configurations. In horizontal well applications, trajectory modifications guided by WMPP values drove measurable gains in cumulative oil production. This methodology concurrently reduced well count requirements and elevated average per-well production volumes, delivering dual economic and operational efficiencies.

In conclusion, the findings from this study underscore WMPP as a powerful and practical tool for reservoir engineers and petroleum industry professionals. By combining advanced computational techniques with detailed reservoir modeling, WMPP provides a reliable and efficient method for optimizing well placement, maximizing oil recovery, and reducing operational costs. This research offers insights to reservoir development and lays the groundwork for future studies and real-world applications in the petroleum industry.

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